

Conjugation is not independent data: a formalized obstruction for Diaz’s modulus question

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Abstract

Let $K \subseteq \mathbb{C}$ be a conjugation-stable subfield and let $u \in \mathbb{C}$ satisfy $u\bar{u} \in K$. We record an elementary observation and its consequences, all machine-checked in Lean 4 with Mathlib: since $\bar{u} = (u\bar{u})/u$, complex conjugation on the subfield generated by K and u is determined by the ring structure rather than being extra data. Three consequences follow. Any ring homomorphism fixing K and sending u to a second point t on the same circle automatically intertwines conjugation on that subfield; such a homomorphism exists on the subfield whenever u and t are transcendental over K ; and no vanishing-coefficient statement over K built from the associated 2×2 matrix distinguishes u from t . These are negative results about a class of proof strategies. No new transcendence result is proved, and Diaz’s question remains open.

1 Setting

Write $\overline{\mathbb{Q}}$ for the field of algebraic numbers in $\mathbb{C}$ and

$$\mathcal{L} = \{u \in \mathbb{C} : e^u \in \overline{\mathbb{Q}}^\times\}$$

for the logarithms of algebraic numbers. Diaz [1] studied the use of complex conjugation in transcendence arguments for the exponential function, asking in particular how the non-holomorphic maps $z \mapsto \bar{z}$ and $z \mapsto |z|$ could enter such a proof at all.

The question we take as our reference point is stated by Diaz [1, §5.1, p. 550] in the following form:

“Soit $u \in \mathbb{C} \setminus \{0\}$ avec $|u| \in \overline{\mathbb{Q}}$; alors e^u est transcendant.”

Equivalently: no non-zero element of $\mathcal{L}$ has algebraic modulus. Diaz notes there that this follows from Schanuel’s conjecture and from the strong four exponentials conjecture; it is stated, credited to [2], at [3, p. 399], and the associated companion condition $|\tau|^2 \in \mathbb{Q}$ is [2, assertion (4-1), p. 237].

Nothing below depends on this attribution. Every result is proved for an arbitrary conjugation-stable subfield $K \subseteq \mathbb{C}$ and is stated that way.

Throughout, $K \subseteq \mathbb{C}$ is a subfield with $\bar{a} \in K$ for every $a \in K$. The case of interest is $K = \overline{\mathbb{Q}}$.

Definition 1.1. For K a subfield of $\mathbb{C}$ and $u \in \mathbb{C}$, let $\text{hull}_K(u)$ denote the subfield of $\mathbb{C}$ generated by K together with u .

Call u a *candidate over K* if $u\bar{u} \in K$; the intended case is $u \in \mathcal{L}$ with $|u|$ algebraic, so that $u\bar{u} = |u|^2 \in \overline{\mathbb{Q}}$.

2 The mechanism

The whole note rests on one identity.

Lemma 2.1. *If $u \neq 0$ and $\rho = u\bar{u}$, then $\bar{u} = \rho/u$.*

Proof. Immediate from $\rho = u\bar{u}$ and $u \neq 0$. $\square$

Trivial as it is, this is the point. For a candidate, ρ lies in K , so $\bar{u}$ is a rational function of u with coefficients in K . Conjugation on $\text{hull}_K(u)$ is therefore not additional structure that a homomorphism might fail to respect; it is already fixed by the field operations.

3 Rigidity

Lemma 3.1. *Let $f, g: \mathbb{C} \rightarrow \mathbb{C}$ be ring homomorphisms agreeing on K and at u . Then $f = g$ on $\text{hull}_K(u)$.*

Theorem 3.2. *Let $u, t \neq 0$ with $u\bar{u} \in K$ and $t\bar{t} = u\bar{u}$. If $\Phi: \mathbb{C} \rightarrow \mathbb{C}$ is a ring homomorphism fixing K pointwise with $\Phi(u) = t$, then*

$$\Phi(\bar{z}) = \overline{\Phi(z)} \quad \text{for every } z \in \text{hull}_K(u).$$

Proof. Both $\Phi \circ \text{conj}$ and $\text{conj} \circ \Phi$ are ring homomorphisms agreeing on K , because K is conjugation-stable and Φ fixes it. At u they agree too: by Lemma 2.1, $\Phi(\bar{u}) = \Phi(\rho/u) = \rho/t$ and $\bar{t} = \rho/t$. Now apply Lemma 3.1. $\square$

The intertwining is not arranged. It is forced.

4 Existence

Theorem 3.2 is conditional on a homomorphism existing. On the hull, one does.

Theorem 4.1. *If u and t are both transcendental over K , there is a K -algebra homomorphism $\Phi: K(u) \rightarrow \mathbb{C}$ with $\Phi(u) = t$.*

Proof. For f transcendental over K , the field $K(f)$ is K -isomorphic to the field of rational functions $K(X)$, by $X \mapsto f$. Applying this to u and to t gives $K(u) \cong K(X) \cong K(t)$ over K ; compose with the inclusion $K(t) \hookrightarrow \mathbb{C}$. $\square$

Remark (scope of Theorem 4.1). The homomorphism is produced on $K(u)$, not on all of $\mathbb{C}$. Extending a homomorphism from $K(u)$ to an endomorphism of $\mathbb{C}$ is possible, by extending a transcendence basis and using that $\mathbb{C}$ is algebraically closed, but it is not proved here and is not needed: every use below evaluates Φ only on $\text{hull}_K(u)$.

5 The obstruction

Attach to a candidate the matrix

$$H(u, r) = \begin{pmatrix} u & r \\ r & \bar{u} \end{pmatrix},$$

where $r \in K$ satisfies $u\bar{u} = r^2$. Then $\det H = u\bar{u} - r^2 = 0$, so H has rank one, and its coefficients factor.

Theorem 5.1. *Let $r \in K$, let $u \notin K$ with $u\bar{u} = r^2$, and let $w, v \in K^2$ be non-zero. Then*

$$\sum_{i,j} w_i H(u, r)_{ij} v_j \neq 0.$$

Corollary 5.2. *Let $r \in K$ and let $u, t \notin K$ both satisfy $u\bar{u} = t\bar{t} = r^2$. Then for all non-zero $w, v \in K^2$,*

$$\sum_{i,j} w_i H(u, r)_{ij} v_j = 0 \iff \sum_{i,j} w_i H(t, r)_{ij} v_j = 0.$$

Proof. By Theorem 5.1 neither side ever holds. □

Corollary 5.2 is the negative statement. A proof strategy that accumulates vanishing-coefficient constraints over K on u and $\bar{u}$, hoping they collide, cannot separate a candidate from an ordinary point placed on the same circle. The reason is Lemma 2.1: the constraints are not independent, because $\bar{u}$ was never independent of u .

6 What is and is not claimed

Claimed: the statements above, for an arbitrary conjugation-stable subfield K , each machine-checked.

Not claimed. No transcendence result, and no progress on whether a non-zero element of $\mathcal{L}$ can have algebraic modulus; that question is untouched and remains open. Nothing here shows the conjugation method fails in general — only that this particular family of statements, vanishing of coefficients over K of the rank-one matrix H , does not separate candidates. Theorem 4.1 is proved on $K(u)$ only. The formalization does not assert that any candidate exists; establishing that a non-zero $u \in \mathcal{L}$ with $|u|$ algebraic is transcendental requires Hermite–Lindemann, which the compared surface of this note deliberately avoids.

7 Related work, and what was searched

Prior art, attributed

Two ingredients used above are not new and are not claimed.

The matrix $H(u, r)$ of §6 is the configuration $x_1 = 1, x_2 = |\lambda|/\lambda, y_1 = \lambda, y_2 = |\lambda|$ recorded at [3, p. 15]. The question itself, and the companion condition $|\tau|^2 \in \mathbb{Q}$, are Diaz’s [1, §5.1, p. 550] and [2, assertion (4-1), p. 237] respectively, the former also at [3, p. 399]. The divisor of labour is therefore: the configuration and the question are Diaz’s and Waldschmidt’s; what is offered here is the observation that Lemma 2.1 makes conjugation on $\text{hull}_K(u)$ redundant data, and the three consequences drawn from it.

The search

The following were examined for the observation of §3 and its consequences — that is, for any statement that conjugation on $\overline{\mathbb{Q}}(u)$ is determined by the ring structure, for any use of a homomorphism of $\mathbb{C}$ fixing $\overline{\mathbb{Q}}$ to transport a hypothetical counterexample, or for any indistinguishability conclusion argued by transport rather than by estimates. Searches were run in English and French, in two independent passes, in August 2026.

- [1] and [2], both read in full. Neither contains any statement of the kind. Both were also swept mechanically for the vocabulary such an argument requires — *homomorphisme*, *endomorphisme*, *isomorphisme*, *automorphisme*, *plongement*, *extension* — and the count is zero in each paper for every term. Neither paper constructs or uses a map of fields.

- [3], at pp. 15, 399 and 614, including Exercise 15.16, together with the conjectures involving modulus and conjugation to which [1, p. 551] directs the reader. Exercise 15.16 does not contain the observation.
- Mantova, *Involutions on Zilber fields* [4], which concerns involutions on abstract exponential fields rather than subfields of $\mathbb{C}$, and Dasgupta–Kakde [5], Conjectures 1.2 and 2.3.
- arXiv full-text search, and general web search, on English and French keyword families covering algebraic modulus of a logarithm, complex conjugation in transcendence, rational function fields with involution, and transfer or indistinguishability obstructions.

Outcome, and the nearest statement

The observation was not located in any source examined.

The nearest thing found is a question rather than a statement. Having derived his conjecture from Schanuel’s, Diaz turns to method [1, p. 551]:

“Comment utiliser l’hypothèse $|u| \in \overline{\mathbb{Q}}$? Comment faire apparaître $|u|$? Peut-on envisager d’utiliser des fonctions comme $z \mapsto \bar{z}$, $z \mapsto |z|$?”

Lemma 2.1 is one answer: it makes the hypothesis bite by converting $u\bar{u} = \rho$ into a statement about the field $\overline{\mathbb{Q}}(u)$. But Diaz asks the question and does not answer it, and nowhere observes that conjugation on $\overline{\mathbb{Q}}(u)$ is determined by the ring operations, which is the whole content here.

As far as the author knows, as far as the search recorded above reached, and as far as private enquiry established, the observation of §3 and the consequences in §§4–6 are not in the literature. Those replies are not reproduced here and the correspondents are not named.

That is an assertion about a negative, made after looking. It is not a proof of priority, and a reader who knows of a prior appearance is asked to say so.

Limits of this search

Stated so the previous paragraph is read at its true weight. The searches were conducted with substantial assistance from large language models, and one of the two passes misread a scanned source at a decisive point; the error was caught by checking the printed page and is corrected above. Text extracted from scanned journal PDFs loses overlines, so $\mathbb{Q}$ and $\overline{\mathbb{Q}}$ are not reliably distinguishable in such sources without consulting page images. Review services behind paywalls were not searched systematically, and monograph literature, much of it French and predating 1995, is poorly covered by full-text search. The private enquiry mentioned above is part of the author’s basis for the claim but is not evidence a reader can audit; the checkable part of the search is the list of public sources, each of which can be consulted at the page cited.

8 Use of AI assistance

The mathematics, the formalization and this note were developed with substantial assistance from large language models, directed and accepted by the author, who is responsible for the content. The Lean development was written with such assistance and is machine-checked at the pinned toolchain; the axiom closure of every statement in the table below was audited with `#print axioms`. The literature search recorded above was likewise AI-assisted, with the limits stated there. No claim in this note rests on an assertion by a language model: every mathematical statement is checked by Lean, and every bibliographic claim is checked against the cited page.

9 Formalization

The development is Lean 4 with Mathlib; sources at <https://github.com/carlok/diaz-modulus-lean>. The advertised statements are in `Challenge.lean`, with proofs in `Solution.lean`.

Statement	Lean declaration	Module
Lemma 2.1	<code>conj_eq_rho_div</code>	<code>Diaz.Closure</code>
Lemma 3.1	<code>eq0n_hull</code>	<code>Diaz.Closure</code>
Theorem 3.2	<code>conj_comm</code>	<code>Diaz.Transfer</code>
Theorem 4.1	<code>exists_algHom_of_transcendental</code>	<code>Diaz.Palomar</code>
Theorem 5.1	<code>no_vanishing_coeff_matrix</code>	<code>Diaz.Rigidity</code>
Corollary 5.2	<code>coeff_indistinguishable</code>	<code>Diaz.Palomar</code>
$\text{hull}_K(u)$	<code>hull</code>	<code>Diaz.Closure</code>
$H(u, r)$	<code>Hmat</code>	<code>Diaz.Rigidity</code>

Running `#print axioms` on each of these lists only `propext`, `Classical.choice` and `Quot.sound`. The wider repository also contains two imported transcendence statements declared as axioms, `hermite_lindemann` and `exists_ringHom_of_transcendental`, used for results outside the list above. No statement in the table depends on either.

References

- [1] G. Diaz, *Utilisation de la conjugaison complexe dans l'étude de la transcendance de valeurs de la fonction exponentielle usuelle*, J. Théor. Nombres Bordeaux **16** (2004), no. 3, 535–553, [doi:10.5802/jtnb.459](https://doi.org/10.5802/jtnb.459).
- [2] G. Diaz, *La conjecture des quatre exponentielles et les conjectures de D. Bertrand sur la fonction modulaire*, J. Théor. Nombres Bordeaux **9** (1997), no. 1, 229–245, [numdam](https://numdam.org/).
- [3] M. Waldschmidt, *Diophantine Approximation on Linear Algebraic Groups*, Grundlehren der mathematischen Wissenschaften **326**, Springer, 2000.
- [4] V. Mantova, *Involutions on Zilber fields*, [arXiv:1109.6155](https://arxiv.org/abs/1109.6155).
- [5] S. Dasgupta and M. Kakde, *On the Brumer–Stark conjecture and refinements*, [arXiv:2408.08178](https://arxiv.org/abs/2408.08178).
- [6] The Mathlib Community, *Mathlib: the mathematical library of Lean 4*, [project repository](https://project-repository.org/).